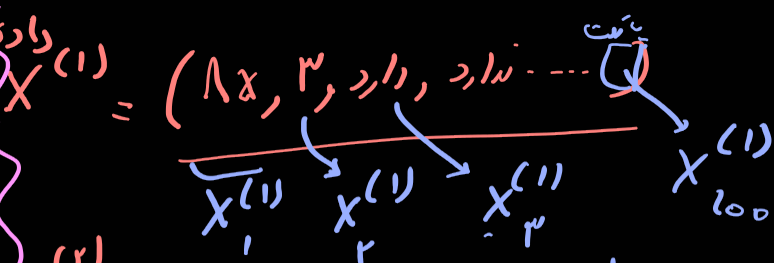




$y^{(1)} = 12$

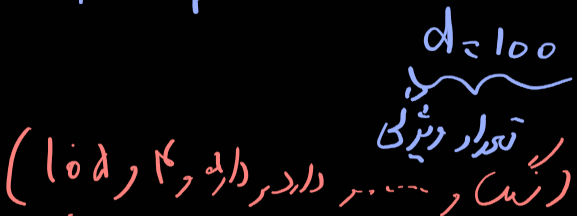


$y^{(1)} = 12$

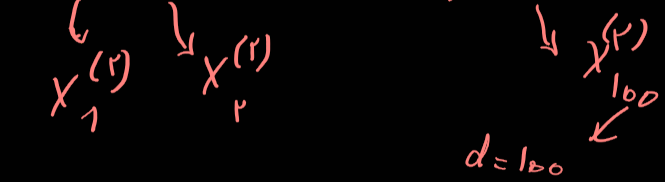
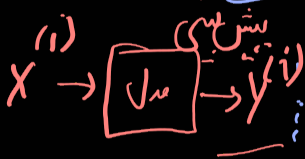


$y^{(2)} = 14$

خبر



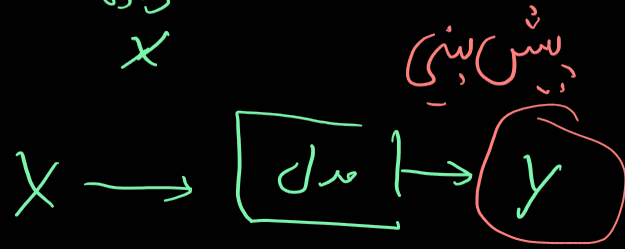
$y^{(2)} = 14$



$y^{(3)} = \dots$

$X^{(1)} \dots X^{(1000)}$ داده ها

$y^{(1)} \dots y^{(1000)}$ برچسب



پیش بینی

برچسب

Generalize

تعمیم پذیری

$$y = \underline{a}x + \underline{b}$$

$x^{(1)}$

$y^{(1)}$

Shuffle

بهرین

Train 70%

۱۰۰٪ استفاده

Validation 10%

۱۰٪ استفاده

Test 20%

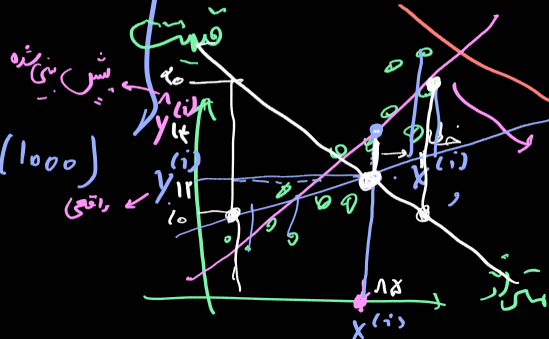
نتایج نهایی

$$y = \textcircled{a}x + \textcircled{b}$$

۱۰۰٪ استفاده

$x^{(1000)}$

$y^{(1000)}$



$1000 \left\{ \begin{array}{ll} 700 & \text{Train} \\ 100 & \text{Validation} \\ 200 & \text{Test} \end{array} \right. \rightarrow G, \text{inc}, b^v$

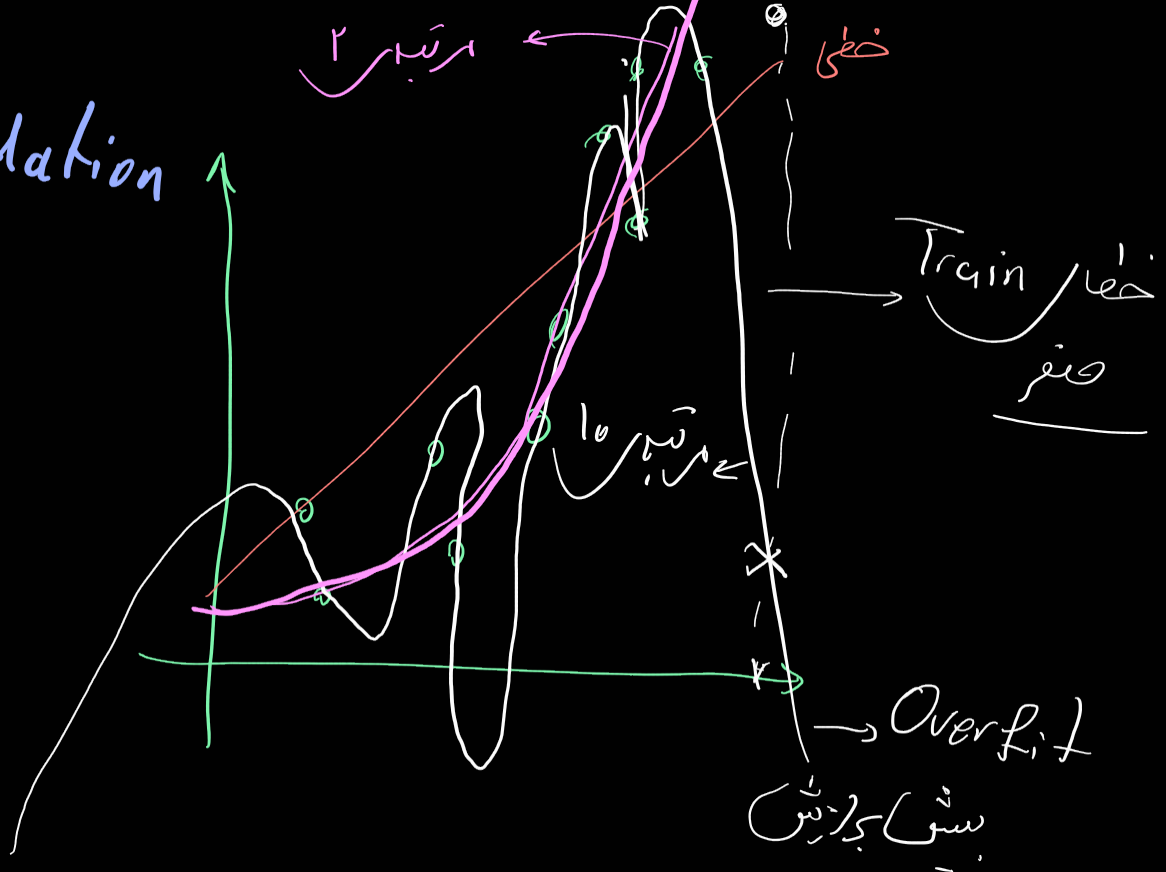
1.
K-fold

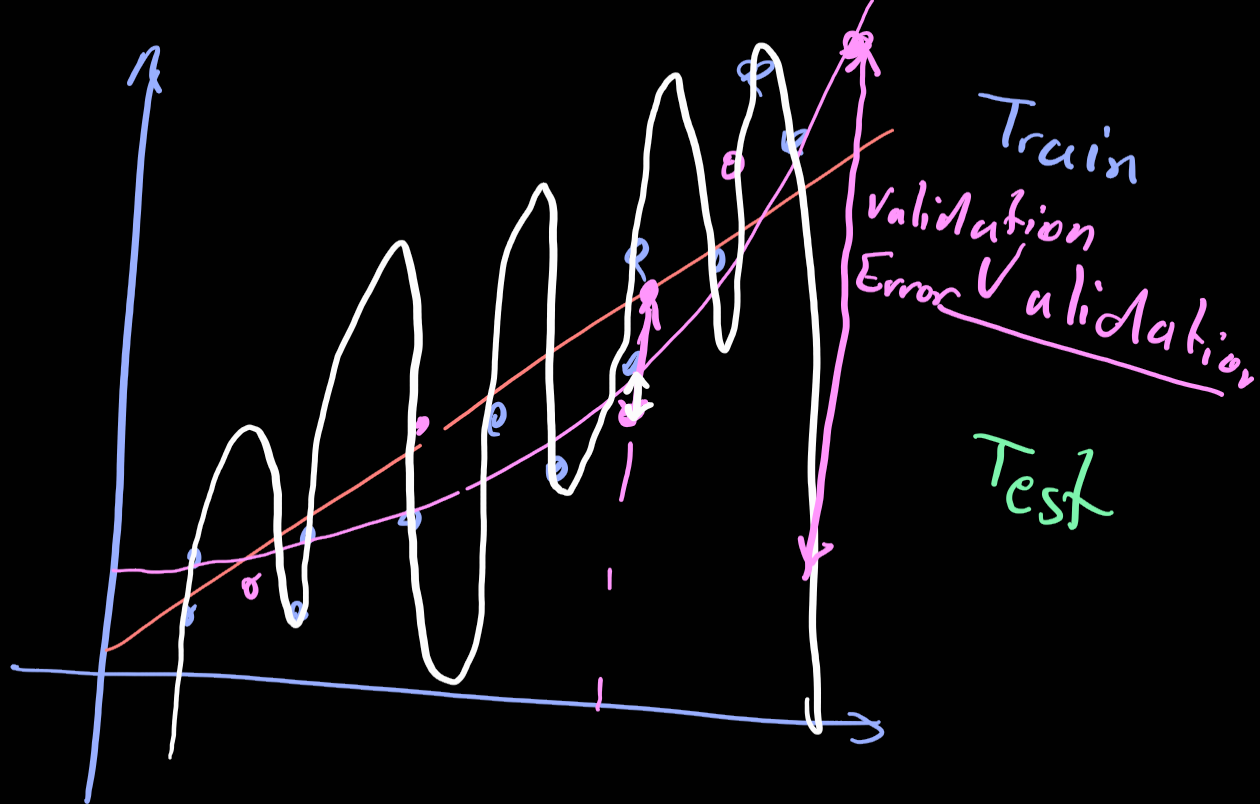
$$700 \rightarrow y = \underbrace{a}_1 n + \underbrace{b}_2 \quad \text{Validation}$$

$$700 \rightarrow y = \underbrace{a}_1 n^r + \underbrace{b}_2 n + \underbrace{c}_3 \rightarrow \text{Validation}$$

$$700 \rightarrow y = a n^r + b n^r + c n + d + \underbrace{e}_5$$

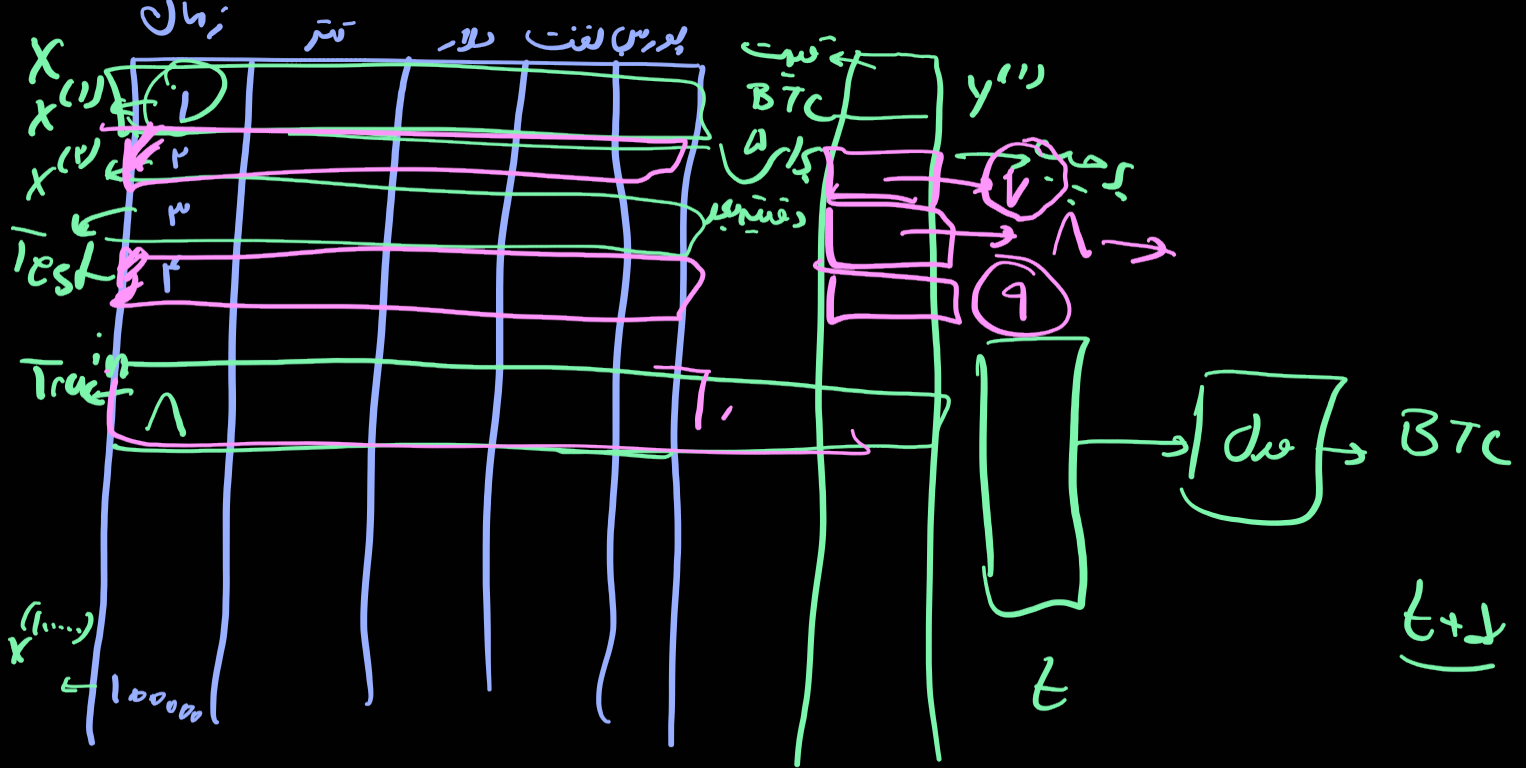
Validation

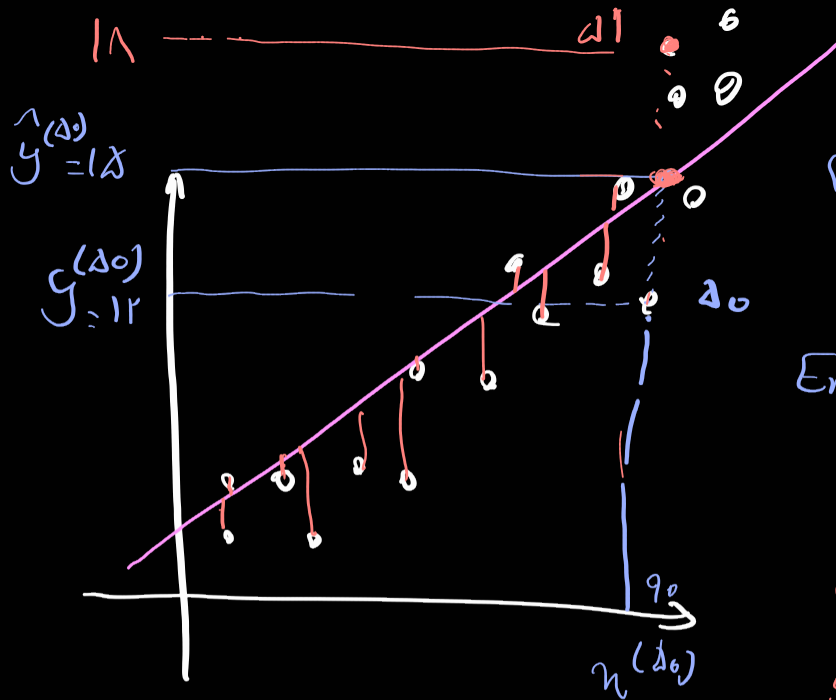




خطای Train	خطای Validation	تعداد
زیاد	زیاد	۱
کم	کم	۲
صفر	خطای زیاد	۱۵

X





$$\text{Error} = \hat{y}^{(i)} - y^{(i)}$$

$$\text{Error}^{(\Delta_0)} = 18 - 12 = 6$$

$$\text{Error}^{(\Delta_1)} = 12 - 18 = -6$$

$$x^{(\Delta_1)} = x_0$$

$$y^{(\Delta_1)} = 18$$

$$\hat{y}^{(\Delta_1)} = 12$$

Loss Function

$$\text{MAE} = \sum_{i=1}^N \frac{|\hat{y}^{(i)} - y^{(i)}|}{N}$$

Mean Absolute Error

$$\text{Mean Squared Error} = \frac{\sum_{i=1}^N (\hat{y}^{(i)} - y^{(i)})^2}{N}$$

MSE